

GEOptimize

167 Lombard Avenue, Suite 500
Winnipeg, MB R3B OV3
+1 (204) 318-2156

Proposal for:	Proposal Date:	Proposal Number:
EMCOR Services Betlem Rochester, NY	2024-04-25	QU-0158
ATTN: Jonathon Suhre, Mechanical Engineer		Reference US NY EMCOR Rochester

Understanding of Project

The City of Rochester is soliciting proposals to develop a district geothermal system in the areas highlighted in the following map of the downtown area. The proposed conceptual plan includes a total of approximately 15 multi-family residential buildings. Buildings proposed at the west end of the area may include some commercial / retail space. Buildings proposed for the central area will primarily residential. The eastern end of the proposed development will include approximately 50 single family residential homes.

The intent is to fill the area above the existing freeway with an average of approximately 20' of fill. It may be possible to construct a ground heat exchanger (GHX) by inter-layering GHX piping under and at various levels as the fill is being placed.



EMCOR has requested assistance from GEOptimize in developing a conceptual design of the geothermal district energy system.

Proposed Design Methodology

The size, cost, and performance of a GHX is sensitive to the peak heat transfer to and from the ground. It is also sensitive to total heat rejection to the ground when the building is being cooled compared to the total heat extraction from the ground to heat the building. This is referred to as the annual energy balance. A large, ongoing annual imbalance of heat rejection versus heat extraction can gradually compromise the efficiency of heat pumps connected to the system as the temperature of the GHX increases or decreases over time. Increasing the size of the GHX only slows long-term temperature degradation but does not eliminate the deterioration of system performance.

The type and configuration of ground heat exchanger (GHX) that will be selected for the district GSHP system will be determined based on a review of the local geology, hydrogeology, project site and drilling space available. Hourly energy loads provided by EMCOR will be used for this analysis. The optimal combination of types of sources and sinks will be determined from using TRNSYS.

TRNSYS software can be used to analyse the relationship between the GHX's, buildings, and other potential energy resources. Other energy sources and / or heat sinks that can be modelled with TRNSYS are wastewater heat transfer, heat dissipation with a snow melt system, heat injection from solar thermal collectors, open water well heat exchangers, pond heat exchangers, heat dissipation from fluid coolers or cooling towers, etc. A series of simulations using TRNSYS will be developed to optimize the system configuration including:

- The size and configuration of sources and sinks.
- Location of GHX modules in relation to building heating and cooling requirements.
- System control strategy.

Information Required

- 8760 hourly energy models for each of the separate buildings including:
 - Space heating and cooling loads post retrofit without the heat pump compressor energy.
 - Domestic hot water loads.
- Thermal conductivity test report for soil thermal properties.

Scope of Work

- Develop representative hourly energy models for 15 multi-family residential buildings approximately 50,000 to 80,000 ft² in area. Approximately half of the buildings will include some commercial space on the main floor level. A Rochester, NY Typical Meteorological Year (TMY) weather file will be used to create the representative energy models.
- Develop hourly energy models for 3 single family homes ranging in size from 1,400 to 2,000 ft². The homes will each be modelled with different solar orientations, using a Rochester, NY TMY file.
- Design of central single pipe system, including pipe sizes and hydronic equipment utilizing TRNSYS.

- EMCOR team to develop routing of ambient loop and supply/return lines.
- Connection design for each building, ambient loop, and heat sources and sinks.
- Pressure-drop calculations for the GHX and circulation pump sizing for ambient loop, heat source and sinks, and building supply/return lines.
- Auxiliary heating and cooling options and selection for conceptual.
- Specifications for heat transfer fluid and GHX piping for circuits, headers, supply / return runout piping up to and including the manifold.
- Coordination with EMCOR for interior mechanical system and heat pump selection.
- Develop long term management and detailed system control strategy ensuring system operates within expected design parameters for the life of the system.

Deliverables

- Single line drawing describing overall system schematic. This includes site plan indicating locations of the GHX and district piping.
- Design brief describing the following:
 - Assumptions used in modelling.
 - Description of the overall system and how it should be operated.
 - GHX descriptions in terms of number of bores, length of bores, pipe sizes, fluid specifications, and pressure drop calculations.
 - System control sequence.
 - Long term management plan and specification of data monitoring package
- Meetings/coordination calls.

Pricing

Description			Line Total
District Detailed Design			\$49,300.00
Total (USD)			\$49,300.00

Authorized signature

General Terms & Conditions for Professional Services (USA)

(a) These terms and conditions of sale (these "**Terms**") are the only terms that govern the sale by **GEOptimize, Inc.** ("**Service Provider**") of the services ("**Services**"), as more particularly described on the Service Provider's Proposal to which these Terms are attached (the "**Proposal**") to the client named on the Proposal ("**Client**"). Service Provider and Client are sometimes referred to in these Terms as a "**Party**" or the "**Parties**."

(b) The accompanying Proposal and these Terms (collectively, this "**Agreement**") comprise the entire agreement between the parties and supersede all prior or contemporaneous understandings, agreements, negotiations, representations and warranties, and communications, both written and oral. These Terms prevail over any of Client's general terms and conditions of purchase regardless of whether or when Client has submitted its purchase order or such terms. Fulfilment of Client's order does not constitute acceptance of any of Client's terms and conditions and does not serve to modify or amend these Terms. Notwithstanding anything herein to the contrary, if a written contract signed by both parties is in existence covering the sale of the Services covered hereby, the terms and conditions of said contract shall prevail to the extent they are inconsistent with these Terms.

2. Performance of Services.

(a) Services. In accordance with this Agreement and applicable law, Service Provider shall provide to Client the Services set forth in the Proposal.

(b) Change Orders; Scope Changes. Any additions to or changes to the scope of Services must be agreed by the parties in a written change order ("**Change Order**") or other writing. The Parties will equitably adjust pricing and schedule and enter into a Change Order reflecting such adjustments if (i) a scope change is necessary due to emergency, safety, or other reasonable necessity, or (ii) if specific periods of time for rendering Services set forth in the Proposal are exceeded through no fault of Service Provider.

(c) Client's Obligations. In connection with the Services, Client shall reasonably cooperate with Service Provider and provide Service Provider with information and access to Client's premises and facilities, as may reasonably be requested by Service Provider.

3. Delays. If Service Provider's performance of its obligations under this Agreement is prevented or delayed by any act or omission of Client or its agents, subcontractors, consultants, or employees ("**Client Caused Delay**"), Service Provider shall not be deemed in breach of its obligations under this Agreement or otherwise liable for any costs, charges, or losses sustained or incurred by Client, in each case, to the extent arising directly or indirectly from such prevention or delay. In addition, Service Provider shall not be deemed in breach of its obligations under this Agreement for any delays to the extent resulting from circumstances or events outside of Service Provider's control.

4. Fees and Expenses.

(a) In General. In consideration of the provision of the Services by the Service Provider and the rights granted to Client under this Agreement, Client shall pay the fees ("**Fees**") as follows:

(i) Time and Materials. For any Services provided on a time and materials basis, as specified in the Proposal, Client shall pay Fees calculated in accordance with Service Provider's daily/hourly fee rates as set forth in the Proposal.

(ii) Fixed Price. Where Services are provided for a fixed price, as specified in the Proposal, the total Fees shall be the amount set out in the Proposal.

(b) Travel Expense Reimbursement. Client shall reimburse Service Provider for any travel necessary to perform the Services as specified in the Proposal or as otherwise approved by Client in writing.

(c) Taxes. All Fees are exclusive of all sales, use, value added and excise taxes, and any other similar taxes, duties, and charges of any kind imposed by any governmental authority on any amounts payable by Client. Client shall be responsible for all such charges, costs, and taxes. To the extent Service Provider is required to pay any such sales, use, excise, value added or other taxes or duties or charges, Client shall reimburse Service Provider for such payments. Notwithstanding the foregoing, Client shall not be responsible for any taxes imposed on, or with respect to, Service Provider's income, revenues, gross receipts, personal or real property, or other assets.

(d) Collection Expenses. If Service Provider incurs any costs, expenses, or fees, including reasonable legal fees and professional collection services fees, in connection with the collection or payment of any amounts due it under this Agreement, Customer agrees to reimburse Service Provider for all such costs, expenses and fees.

5. Payment Terms. Unless otherwise specified in the Proposal, Client shall pay Fees within 30 days of an invoice from Service Provider. Invoices that remain unpaid for more than 15 days after the payment due date may, at Service Provider's discretion, bear interest at the rate of one percent (1%) per month or the maximum rate permitted by law, whichever is less, calculated daily and compounded monthly. Client shall also reimburse Service Provider for all costs it incurs in collecting any late payments, including, without limitation, reasonable attorneys' fees and court costs. In addition to all other remedies available under the Agreement or at law, Service Provider shall be entitled to suspend the performance of any Services if Client fails to pay any amounts when due hereunder and such failure continues for 30 day following written notice thereof, and such suspension of Services shall not be considered a breach or default of any of Service Provider's obligations under this Agreement.

6. Limited Warranty.

(a) Service Provider warrants that it shall provide the Services (a) in accordance with the terms of this Agreement; (b) using personnel of industry standard skill and experience who are qualified and licensed, as required; and (c) in a timely, workmanlike, and professional manner in accordance with generally recognized industry standards for similar services.

(b) Except for the warranties set forth in section 6(a), Service Provider makes no warranty whatsoever with respect to goods or services, including any (a) warranty of merchantability; (b) warranty of fitness for a particular purpose; or (c) warranty against infringement of intellectual property rights of a third party; whether express or implied by law. Service Provider shall not be liable for any services or work product or deliverables provided by third-party vendors identified or referred to the Client by the Service Provider during the term of this Agreement, pursuant to any Proposal or otherwise, except if such third-party services are provided under written agreement with Service Provider.

(c) Service Provider shall not be liable for a breach of the warranties set forth in Section 6(a) unless (i) Client gives written notice of the defective Services, reasonably described, to Service Provider within 90 days after the completion of the Services and (ii) Service Provider reasonably verifies Client's claim that the Services are defective.

(d) With respect to any Services subject to a claim under the warranty set forth in Section 6(a), Service Provider shall, in its sole discretion, (i) repair or re-perform the applicable Services, without additional charge or (ii) credit or refund the price of such Services attributable to the applicable Services.

(e) The remedies set forth in section 6(d) shall be the Client's sole and exclusive remedy and Service Provider's entire liability for any breach of the limited warranty set forth in section 6(a).

7. Limitation of Liability.

(a) **In no event shall Service Provider be liable to Client or any third party for any consequential, indirect, incidental, special, exemplary, or punitive damages whether arising out of breach of contract, tort (including negligence), strict liability or otherwise, regardless of whether such damages were foreseeable and whether or not Service Provider has been advised of the possibility of such damages, and notwithstanding the failure of any agreed or other remedy of its essential purpose.**

(b) **In no event shall Service Provider's aggregate liability arising out of or related to this agreement, whether arising out of or related to breach of contract, tort (including negligence), strict liability or otherwise, exceed the total of the amounts paid to Service Provider for the services provided under this Agreement.**

8. Intellectual Property. All intellectual property rights, including copyrights, patents, patent disclosures and inventions (whether patentable or not), trademarks, service marks, trade secrets, know-how and other confidential information, trade dress, trade names, logos, corporate names and domain names, together with all of the goodwill associated therewith, derivative works and all other rights (collectively, "**Intellectual Property Rights**") in and to all documents, work product and other materials that are delivered to Client under this Agreement or prepared by or on behalf of the Service Provider in the course of performing the Services, including any items identified as such in the Proposal (collectively, the "**Deliverables**"), except for any Confidential Information of Client or Client materials, shall be the sole and exclusive property of the Service Provider. Subject to Client's payment of the Fees, Service Provider hereby grants Client, a license to use all Intellectual Property Rights in the Deliverables free of additional charge and on a non-exclusive, worldwide, non-transferable, non-sublicensable, fully paid-up, royalty-free and perpetual basis to the extent necessary to enable Client to make reasonable use of the Deliverables and the Services. The Client shall indemnify, defend and hold Service Provider harmless from and against all claims, actions or suits arising out of or related to the Client's unauthorized use of the Deliverables or infringement upon or misappropriation of Service Provider's Intellectual Property Rights.

9. Term; Termination and Survival.

(a) This Agreement shall commence as of the Effective Date and shall continue thereafter until the completion of the Services (the "**Term**") unless sooner terminated pursuant to paragraphs (b) or (c) of this Section 9.

(b) Either Party may terminate this Agreement, effective upon written notice to the other Party (the "**Defaulting Party**") if the Defaulting Party:

(i) breaches this Agreement (other than a payment default governed by paragraph (c) of this Section), and the Defaulting Party does not cure such breach within 30 days after receipt of written notice of such breach, or such breach is incapable of cure; or

(ii) experiences a bankruptcy, insolvency, or financial insecurity event in which it: (i) becomes or is declared insolvent or bankrupt; (ii) is the subject of a voluntary or involuntary bankruptcy or other proceeding related to its liquidation or solvency, which proceeding is not dismissed within 10 calendar days after its filing; (iii) ceases to do business in the normal course; or (iv) makes an assignment for the benefit of creditors.

(c) Client shall pay Service Provider for all Services rendered prior to the effective date of termination.

10. Waiver. No waiver of any term or right in this Agreement shall be effective unless in writing, signed by an authorized representative of the waiving Party. The failure of either Party to enforce any provision of this Agreement shall not be construed as a waiver or modification of such provision, or impairment of its right to enforce such provision or any other provision of this Agreement thereafter.

11. Confidential Information. From time to time during the Term of this Agreement, either Party (as the "**Disclosing Party**") may disclose or make available to the other Party (as the "**Receiving Party**"), non-public, proprietary, and confidential information of Disclosing Party ("**Confidential Information**"); provided, however, that Confidential Information does not include any information that: (a) is or becomes generally available to the public other than as a result of Receiving Party's breach; (b) is or becomes available to the Receiving Party on a non-confidential basis from a third-party source, provided that such third party is not and was not prohibited from disclosing such Confidential Information; (c) was in Receiving Party's possession prior to Disclosing Party's disclosure hereunder; or (d) was or is independently developed by Receiving Party without using any Confidential Information. The Receiving Party shall: (x) protect and safeguard the confidentiality of the Disclosing Party's Confidential Information with at least the same degree of care as the Receiving Party would protect its own Confidential Information, but in no event with less than a commercially reasonable degree of care; (y) not use the Disclosing Party's Confidential Information, or permit it to be accessed or used, for any purpose other than to exercise its rights or perform its obligations under this Agreement; and (z) not disclose any such Confidential Information to any person or entity, except to the affiliated entities, directors, officers, members, managers, employees, contractors, agents, advisors, consultants, financing parties, tax equity investors, or lenders of Receiving Party or its affiliated entities ("**Representatives**") who need to know the Confidential Information to assist the Receiving Party, or act on its behalf, to exercise its rights or perform its obligations under this Agreement. The Receiving Party agrees that any of its Representatives to whom Confidential Information is disclosed will be informed of the confidential nature of the information and the Receiving Party's confidentiality obligations under this Agreement, and will be subject to a contractual or other obligation to protect the Confidential Information upon terms consistent with the requirements of this Section. Each Party agrees to be responsible for any act or omission by its Representatives to whom it has disclosed Confidential Information under this Agreement if such act or omission, if done by the Receiving Party, would constitute a breach of this Agreement. If the Receiving Party is required by applicable law or legal process to disclose any Confidential Information, it shall, prior to making such disclosure, use commercially reasonable efforts to notify Disclosing Party of such requirements to afford Disclosing Party the opportunity to seek, at Disclosing Party's sole cost and expense, a protective order or other remedy. The Parties agree that they each shall be entitled to, in addition to other available remedies, to seek specific performance and further injunctive relief to prevent or restrain breaches or threatened breaches of this Section. The confidentiality obligations set forth in this Agreement shall continue for a period of two (2) years after the completion of the Services. The Receiving Party shall not decompile, disassemble, or reverse engineer all or any part of the Confidential Information.

12. Subcontracting. Service Provider shall be responsible for Services performed by any subcontractor. Service Provider may subcontract any part of its obligations under the Agreement.

13. Assignment. Neither Party shall assign any of its rights or delegate any of its obligations under this Agreement without the prior written consent of the other Party (not to be unreasonably withheld), provided however, Service Provider shall have the right to freely assign this Agreement to an affiliated entity or to a successor in interest in connection with a merger, acquisition, or sale of all or substantially all of the assets of Service Provider, or other similar strategic transaction affecting Service Provider. Upon any such assignment, Service Provider will provide prompt written notice to Client. Any purported assignment or delegation in violation of this Section is null and void.

14. Relationship of the Parties. With respect to any Services provided under this Agreement, the Service Provider is an independent contractor and not an employee of the Client. Nothing contained in this Agreement shall be construed as creating any agency, partnership, joint venture, or other form of joint enterprise, employment, or fiduciary relationship between the Parties, and neither Party shall have authority to contract for or bind the other party in any manner whatsoever.

15. No Third-Party Beneficiaries. This Agreement is for the sole benefit of the Parties hereto and their respective successors and permitted assigns and nothing herein, express or implied, is intended to or shall confer upon any other person or entity any legal or equitable right, benefit, or remedy of any nature whatsoever under or by reason of the Agreement.

16. Governing Law and Jurisdiction. This Agreement shall be governed by and construed in accordance with the laws of the State of New York and any federal or state court of competent jurisdiction in New York County, NY shall have jurisdiction to adjudicate any matter arising out of this Agreement. Both Parties hereby submit to the exclusive jurisdiction and venue of any such court.

17. Severability. If any term or provision of this Agreement is invalid, illegal, or unenforceable in any jurisdiction, such invalidity, illegality, or unenforceability shall not affect any other term or provision of this Agreement or invalidate or render unenforceable such term or provision in any other jurisdiction.

18. Survival. Each term and provision of this Agreement that should by its sense and context survive any termination or expiration of this Agreement, shall so survive regardless of the cause and even if resulting from the material breach of either Party to this Agreement.

19. Amendment and Modification. This Agreement, and any exhibits attached hereto, is the entire agreement between the Parties with respect to the subject matter hereof and supersedes any prior agreement or communications between the Parties, whether written, oral, electronic or otherwise. These Terms may only be amended or modified in a writing stating specifically that it amends these Terms and is signed by an authorized representative of each party.

20. Rights Cumulative. The rights and remedies of the Parties herein provided shall be cumulative and not exclusive of any rights or remedies provided by law or equity.

21. Acceptance of Terms. Client shall be deemed to have accepted these Terms if (a) Client signs and returns the Proposal to Service Provider, or (b) Client accepts the performance of Services from Service Provider after Client has had a reasonable opportunity to review the Proposal and these Terms, whichever occurs first.